

A Structured Review of AI-Based Zero-Day Network Intrusion Detection: Models, Evaluation Protocols, Datasets, and Research Challenges

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Abstract - The growing complexity and dynamism of cyber attacks lead to problems with detection of such attacks using NIDS. Particularly, detection of new attacks, which are not present in the training set, is a challenging task. This review systematically examines AI-based zero-day network intrusion detection across four key areas: detection models, datasets, evaluation protocols, and research challenges. We analyze 20 recent studies and find that the choice of evaluation method significantly affects reported performance. Many works employ random train-test splits that overestimate zero-day generalization, while attack-family holdout, temporal, and cross-dataset evaluations provide more realistic assessments. Supervised models remain dominant, with growing interest in anomaly-based, hybrid, few-shot, and transformer-based architectures. Commonly used datasets (NSL-KDD, UNSW-NB15, CICIDS2017) exhibit limitations in traffic realism and attack diversity. Key research gaps include inconsistent definitions of zero-day detection, insufficient cross-dataset evaluation, limited explainability, and lack of deployment-aware assessment. This review provides a structured synthesis to guide future research toward evaluation-centric methodologies for reliable zero-day NIDS.

Keywords - Network Intrusion Detection, Zero-Day Attack Detection, Deep Learning, Machine Learning, Unseen Attacks, CICIDS2017, Anomaly Detection, Generalization, Intrusion Detection Systems.

I. INTRODUCTION

Since the use of interconnected computer networks, cloud systems, Internet of Things (IoT) devices, and digital infrastructure has increased, the attack surface of today's computing systems has also increased. Network Intrusion Detection Systems (NIDS) have thus become an important mechanism in cyber security. NIDS work by observing network traffic and detecting suspicious patterns associated with cyber attacks. Traditional rule-based or signature-based NIDS are based on predefined patterns and are effective for known attacks, but they cannot detect new, unknown, or zero-day attacks effectively. Zero-day attacks are exploitations of previously unknown vulnerabilities or novel attack types not seen during training.

Recently, machine learning (ML) and deep learning (DL) have emerged as two essential techniques in NIDS due to their capability to learn from network traffic. Researchers have proposed a wide range of ML and DL models, including Random Forests, SVMs, CNNs, LSTMs, GRUs, Transformers, and autoencoders. However, despite high accuracy on standard

benchmarks, the generalization of these models to new, unseen, or zero-day attacks remains an open problem.

One of the significant problems of comparison of these works is that there are no consistent criteria for their evaluation. In some works, a random split is used when the same set of attack categories is distributed across training and test sets. In other works, a holdout evaluation of attack families is performed to simulate zero-day conditions. These differences make it extremely difficult to compare the performance of models across different papers.

The selection of datasets for NIDS research also affects the reported outcomes. The datasets commonly used for evaluation of NIDS based on AI techniques include CICIDS2017, NSL-KDD, UNSW-NB15, CSE-CIC-IDS2018, and CICIOT2023. Each dataset differs in terms of the distribution of traffic, attack categories, and characteristics of the data, which affect in turn the observed performance values and the degree to which they can be compared across studies.

Thus, this paper offers a systematic review of AI-based zero-day NIDS research, with focus on four issues: methods, datasets, evaluation procedures, and research challenges. Rather than developing a new model, the goal is to analyze systematically the methodological choices that drive the reported results and identify the research gaps that require attention.

The key goals of this review include:

- Categorizing major approaches based on artificial intelligence that have been applied in zero-day network intrusion detection.
- Comparing the datasets and evaluation protocols employed in recent works.
- Analyzing the differences between traditional evaluation protocols and those designed for zero-days.
- Identifying common research challenges associated with datasets, generalization, explainability, and implementation.
- Summarizing future research directions for reliable zero-day NIDS evaluation.

The rest of this paper is structured as follows. Section II provides background and relevant literature. Section III introduces the structured literature review methodology. Section IV categorizes AI-based detection methods. Section V discusses commonly used datasets and evaluation protocols. Section VI presents a comparative analysis and critical synthesis. Section VII describes research gaps and future directions. Section VIII concludes the paper.

II. BACKGROUND AND RELATED WORK

A. AI-Based Network Intrusion Detection

The machine learning-based NIDS usually learns the relationship between the network traffic properties and attack labels. In classical approaches, Random Forest, SVMs, logistic regression, Naïve Bayes classifier and others are used. The deep learning approach extends this ability by learning high-level representations of network traffic. CNN-based approaches are usually used for learning local patterns in structured or transformed network traffic, while RNN models like LSTM and GRU can learn sequential dependencies. Hybrids of different representation models are possible as well. For example, Bamber et al. explored the use of CNN-LSTM models for cyber intrusion detection on NSL-KDD [4]. The investigation of CNN-GRU architectures for the problem of intrusion detection was done and the combination of convolutional and recurrent feature learning is already an established research topic [5]. The development of recent research also tends towards the use of Transformer-based architectures. Umer et al. explored the wrapper-based feature-selection approach combined with multi-head attention Transformer for intrusion detection using UNSW-NB15 dataset [11]. Systematic reviews of DL-based NIDS similarly report the increasing use of CNNs, recurrent networks, autoencoders, hybrid models, and attention-based architectures [1], [2].

B. Zero-Day and Unseen Attack Detection

Zero-day detection is a challenging task in comparison to traditional classification since the algorithm needs to identify attacks that were not covered in the learning process. Anomaly detection is one way of addressing the issue by teaching the model how the normal or known traffic looks like and detecting deviations from it. Dai et al. studied autoencoders for zero-day detection in the context of CIC-MalMem-2022 dataset and used learned features with Random Forest and XGBoost classifiers [6]. Another option is to exclude certain types of attacks from the training process and test the algorithm on those attacks that were unseen. Family-holdout evaluation is a more explicit way of assessing the ability of the model to generalize the attacks. The recent studies explored zero-day behavior and traffic drift using Leave-One-Attack-Out-like protocols [7]. Hybrid solutions were also suggested. Hore et al. proposed sequential deep-learning framework incorporating both classification and anomaly-detection components for known, zero-day, out-of-distribution

and adversarial traffic [8]. More recently, Park and Lim proposed a multi-stage framework combining clustering, a variational autoencoder, and supervised deep learning for identifying zero-day and known attacks using CICIDS2017 and CICIDS2018 [12]. Another new research direction includes meta-learning. Ijaz et al. have investigated Model-Agnostic Meta-Learning and Prototypical Networks for zero-day detection in case of data scarcity on CICIDS2017, CICIOT2023, and CIC-UNSW-NB15 [13] datasets. This line of research attempts to achieve more efficient adaptation to unknown attacks with limited examples.

C. Feature Selection and Representation Analysis

Network traffic datasets can contain many features, including redundant or weakly informative variables. Feature selection is therefore commonly used to reduce dimensionality and computational cost. Otokwala et al. proposed an optimized common feature-selection and deep-autoencoder approach for lightweight IoT intrusion detection and evaluated it using MQTT-IoT-IDS2020 and CICIDS2017 [14]. The study illustrates how feature selection and representation compression can be combined with lightweight deployment.

Feature-selection methods are also being combined with newer architectures. Umer et al. used wrapper-based feature selection before multi-head attention-based Transformer classification [11]. These studies suggest that feature engineering remains relevant even as increasingly complex neural architectures are introduced.

D. Explainability in AI-Based NIDS

Another emerging direction is meta-learning. Ijaz et al. conducted a comparative analysis of Model-Agnostic Meta-Learning and Prototypical Networks for zero-day detection under data scarcity conditions based on CICIDS2017, CICIOT2023, and CIC-UNSW-NB15 [13] datasets. These approaches aim to enhance adaptation to new attack types under data scarcity conditions. Alongside the rise in the usage of complicated ML and DL models, there arose a need to explain the reason behind an IDS decision. Techniques for Explainable Artificial Intelligence (XAI) such as SHAP and LIME gained significance for that purpose. IEEE Access journal published a paper analyzing black-box XAI techniques for network IDS using several datasets and AI models while analyzing explanation quality based on such criteria as descriptive accuracy, sparsity, stability, efficiency, robustness, and completeness [15]. A 2025 survey analyzed XAI

techniques for IDS and discussed related issues of interpretability, evaluation, and application [16]. More recent research continued this line of investigation. A 2026 survey classified XAI techniques for IDS in different scenarios (cloud, IoT, industrial, SDN, and others) and outlined related issues of scalability, adversarial robustness, realistic datasets, and human-in-the-loop operation [17].

III. REVIEW METHODOLOGY

The current literature review follows the methodology of a structured literature review of AI approaches for zero-day network intrusion detection, specifically focusing on models' types, dataset, testing strategy, and research challenges.

A. Research Questions

The review is guided by five research questions.

- RQ1: What artificial intelligence and deep-learning approaches have been investigated for detecting zero-day or previously unseen network attacks?
- RQ2: Which datasets and traffic environments are commonly used to evaluate AI-based zero-day intrusion detection methods, and what limitations affect their suitability for this purpose?
- RQ3: How do existing studies define and evaluate zero-day or unseen attacks, particularly with respect to random train-test splitting, attack-family holdout, temporal evaluation, and cross-dataset evaluation?
- RQ4: What methodological limitations and evaluation biases affect the reported generalization performance of AI-based zero-day NIDS?
- RQ5: What research directions can improve the generalization, adaptability, explainability, and deployment reliability of AI-based zero-day intrusion detection systems?

These questions are designed to shift the review from a model-centered comparison toward an evaluation-centered analysis. This distinction is important because high performance under a conventional train-test split may not reflect actual capability to detect unseen attacks.

B. Literature Search Strategy

Literature relevant to this research topic has been collected based on the search performed by peer-reviewed scientific literature pertaining to network

intrusion detection, zero-day or unseen attack detection, and AI-based security systems. Google Scholar, IEEE Xplore, and Scopus were the primary search databases. The search terms were combinations of such terms as 'network intrusion detection', 'zero-day attack', 'unseen attack detection', 'anomaly-based NIDS', 'machine learning IDS', and 'deep learning IDS'. Titles, abstracts, and full texts of papers were examined for relevance to the research questions.

C. Inclusion and Exclusion Criteria

Studies were included when they met the following criteria:

- The study examined network intrusion detection or a task similar to it in nature.
- The study utilized or evaluated machine learning, deep learning, artificial intelligence, anomaly detection, or other learning-based techniques.
- The study evaluated zero-day, unseen, novel, anomalous, or dynamic attack behavior, or presented methodological evidence regarding such behavior.
- The study adequately described its dataset, evaluation framework, or experiment design to reveal how detection performance was evaluated.
- Published works in peer-reviewed journals and reputable conferences were preferred.
- Recent research from 2024–2026 was selected where applicable; however, older studies that introduced well-known datasets, evaluation concepts, or methodologies were kept.

Studies that concentrated solely on areas other than network intrusion were excluded unless their methodology provided useful evidence toward zero-day network attack detection. Also, studies that failed to provide sufficient detail about their evaluation procedure were excluded.

No study was excluded solely because it reported poor performance. In particular, studies revealing limitations and poor performance under unseen attacks are valuable to the review as they show evidence of the generalization problem.

D. Study Classification and Data Extraction

Classification was done in accordance with both the detection and evaluation methodologies applied in the selected literature.

This extraction methodology stresses the importance of the context in which performance is achieved and does not consider performance values in the papers to

be directly comparable. Indeed, it is necessary to be aware that the same model can present very different performance figures depending on the evaluation protocol: a random split is likely to give higher performance compared to attack-family holdout evaluation.

- In terms of model detection method, there are the following types of literature: supervised classification; anomaly-based detection; hybrid methods; active and adaptive learning; few-shot and meta-learning; transformer and representation-based methods.
- In terms of evaluation method, there are the following types of literature: traditional random train-test evaluation; attack family or attack category hold-out; chronological or temporal evaluation; cross-dataset evaluation; a combination of these evaluation methods.

This extraction methodology stresses the importance of the context in which performance is achieved and does not consider performance values in the papers to be directly comparable. Indeed, it is necessary to be aware that the same model can present very different performance figures depending on the evaluation protocol: a random split is likely to give higher performance compared to attack-family holdout evaluation.

E. Synthesis Strategy

The set of studies is qualitatively synthesized by comparing model assumptions, datasets, definitions of unseen attacks, and evaluation procedures of the reviewed papers. It is not possible to synthesize quantitatively because heterogeneous evaluation protocols make metric comparison meaningless.

The synthesis of papers proceeds in three stages. First, the review presents a classification of AI-based detection methods and analyzes how various types of models are used for detecting both known and zero-day attacks. Second, it describes the most common datasets and evaluation frameworks and critically evaluates whether these datasets and protocols allow testing of zero-day generalization. Third, it links these observations to challenges and limitations, such as the absence of standard definitions, dataset-dependent results, and real-world deployment concerns.

The resulting synthesis is therefore organized around what is being detected, how “unseen” behavior is defined, how the evaluation is performed, and what conclusions can reasonably be drawn from the results.

IV. TAXONOMY OF AI-BASED ZERO-DAY INTRUSION DETECTION

Zero-day attack identification using AI-based systems may include several learning strategies that are distinguished by their approaches to modeling of known attacks, normal traffic, incorporation of unlabeled data, and adaptation to novel patterns.

A. Supervised Classification Approaches

Supervised learning approaches including Random Forest, SVM, CNN, LSTM, GRU, DNN, and CNN-LSTM learn decision boundaries on labeled traffic [1], [2], [9]. They are successful as long as the classes of attack do not change too significantly from the training period. Given the unseen attack types, they have inherent limitations to generalization. However, some recent techniques partially circumvent this limitation through representation learning and generalized feature extraction, though the evaluation method needs to be able to test for generalization beyond the standard split.

B. Anomaly-Based Detection

The idea of anomaly-based approaches is to understand what normal network traffic looks like and detect abnormal instances. Methods such as autoencoders, deep autoencoders, reconstruction and novelty scoring techniques can detect deviations from learned normal patterns [3], [6]. These have a theoretical advantage for zero-day detection in that they do not require labeled attack examples; however, they may experience high false-positive rates or fail to detect subtle anomalies.

C. Hybrid Detection Approaches

Hybrid techniques utilize complementary techniques like classifier-based systems with anomaly detectors, CNNs with RNNs, or feature selection techniques with deep learning models [1], [2], [4]. The promise of these techniques is higher robustness and better generalization. Hore et al. and Park and Lim built hybrid frameworks that integrated multiple detection strategies [8], [12].

D. Adaptive and Active Learning

Adaptive, continual, and active learning techniques adapt to changes in network traffic based on new data or feedback [13], [19]. These techniques can prove useful whenever there is new data for attack types or when drift of traffic is detected. Wu et al. used a deep Q-network to implement active learning for zero-day detection [19]. Such approaches may be especially valuable when labeled data for new attacks becomes available incrementally.

E. Few-Shot and Meta-Learning

Few-shot learning seeks to detect new attack classes from a few labeled samples, whereas meta-learning involves transferring knowledge acquired through multiple tasks to new tasks [16], [18]. This is relevant in the zero-day context because few or no examples may be available for a new attack. Ijaz et al. applied Model-Agnostic Meta-Learning and Prototypical Networks to this problem [13]. These approaches show promise but remain at an early stage of development for NIDS.

F. Transformer and Representation-Based Approaches

Transformer-based and representation-learning models make use of attention models, fusion of features, and learned representations for modeling complex relationships in network traffic [6], [18]. Self-supervised and contrastive learning methods also fall into this category. Such techniques hold the potential to create general representations that could transfer more easily to unseen attack types, though their performance on holdout and cross-dataset evaluations needs further investigation.

G. Comparative Synthesis

These six types of classification techniques address specific requirements for zero-day intrusion detection. For supervised techniques, it is necessary to have strongly discriminating power, whereas for anomaly-based systems, it is needed to have good calibration of the normal traffic distribution. Hybrid approaches provide a balance between classification precision and anomaly-based coverage. Adaptive and few-shot methods address the challenge of ongoing change in the attack landscape. Transformer-based approaches try to provide rich representation. The choice among these types affects not only the detection performance but also what form of evaluation is appropriate.

V. DATASETS AND EVALUATION PROTOCOLS

The estimation of zero-day intrusion detection significantly depends on the selected dataset and train/test division technique. Well-known NIDS datasets like NSL-KDD, UNSW-NB15, and CICIDS2017 include a variety of attack classes, but their traffic characteristics, attack compositions, and real-world representativeness vary considerably.

A. Common NIDS Datasets

NIDS datasets vary in terms of attack diversity, traffic distribution and realism. While traditional datasets are

still valuable for comparative analysis, their limitations can play role in zero-day evaluation. NSL-KDD is derived from KDD Cup 1999 and remains widely used for benchmark comparison despite being old and having some unrealistic traffic characteristics. UNSW-NB15 offers a more recent mix of modern attack types and normal traffic. CICIDS2017 is widely used due to its variety of attack families and realistic network simulation. However, various researchers have identified issues with duplicate records, artificial labeling artifacts, or restricted attack diversity in these datasets [10]. CICIoT2023 was introduced specifically for IoT contexts and CIC-MalMem-2022 addresses malware-based attacks. These newer datasets extend evaluation but remain limited in temporal diversity.

B. Random Train-Test Evaluation

Random train/test splitting is traditionally used for evaluation of classical classification performance. But training and test subsets can include samples of the same attack categories and similar traffic patterns, so the model is not required to generalize to novel categories. This leads to inflated results for zero-day scenarios since models are evaluated on a distribution very similar to training. While random split is useful for measuring closed-set classification accuracy, it should not be considered sufficient for evaluating zero-day generalization [9].

C. Attack-Family Holdout Evaluation

The evaluation technique of attack-family holdout omits training one or more attack families and testing the trained model on those omitted families. LOAO evaluation and other related methods offer an explicit way of testing generalization to novel attack categories [7], [14]. This approach is closer to real zero-day scenarios and reveals models that overfit to seen categories. The number and choice of held-out families can affect results substantially.

D. Temporal Evaluation

In case of temporal evaluation, traffic datasets for training and testing are split temporally. This may help to detect changes in traffic distributions and concept drift in real-life cases. However, the temporal nature of most publicly available NIDS datasets is often poorly documented, making rigorous temporal evaluation difficult. Rushendra et al. evaluated temporal stability of an IDS explicitly [7], showing that temporal drift is an important but underexplored concern.

E. Cross-Dataset Evaluation

Cross-dataset evaluation trains a model on one dataset and tests it on another one. This is a good way to measure robustness of models to changes in traffic and attack distributions. However, feature mismatch between datasets is a challenge and feature harmonization is frequently required. Despite the informational value, cross-dataset evaluation is rarely used in practice due to the difficulty of dataset alignment.

F. Evaluation Metrics

Accuracy, precision, recall, F1-score, and false-positive rate are often used for NIDS performance estimation. Accuracy can be misleading when there is class imbalance, and recall is especially valuable for detecting new attacks, whereas false-positive rate indicates the cost of false positives in practice. It is important to interpret performance metrics within the context of the evaluation protocol; a good F1-score obtained via closed-set random split means a different generalization scenario compared to attack-family holdout. In any case, the concept of zero-day performance is inherently dependent on the evaluation procedure. Random split mainly assesses closed-set discrimination, while attack-family holdout directly tests detection of new classes. Temporal and cross-dataset evaluation tackle different kinds of distribution shift and transfer. Thus, it is always necessary to interpret the performance metrics along with the construction of training and testing scenarios.

VI. COMPARATIVE ANALYSIS AND CRITICAL SYNTHESIS

The literature shows that zero-day intrusion detection depends not only on the learning algorithm but also on how zero-day detection is defined, how data are prepared, and how generalization is evaluated.

A. Inconsistent Definition of Zero-Day Detection

Zero-day detection is used to refer to various settings, such as completely unseen attack classes, unknown or anomalous traffic, few-shot adaptation, and traffic change over time [13], [15], [16]. All of these settings have a different amount of information about the unseen attack. Hence, studies must specify the available information during training and testing phases when claiming zero-day capabilities.

B. Dataset Dependence and Evaluation Bias

Dataset properties have an impact on results. NIDS datasets vary in traffic distributions, attack compositions, collection environments, and quality.

Limitations related to dataset, duplicates or similar records, and specific attack-generation procedures influence reported results. Evaluation bias arises when random splits are used instead of attack-family holdout or temporal evaluation [9], [10]. In studies where the training and test sets overlap in attack type, results are likely to reflect closed-set discrimination rather than zero-day generalization.

C. Generalization Across Attack Families

Testing the ability of recognizing attack families that were not seen during training through attack-family holdout and LOAO evaluation is a controllable way of testing models' generalization [14], [17]. Studies that use these protocols often show lower performance than random-split evaluations, highlighting the gap between standard benchmark performance and actual zero-day capability. Sarhan et al. showed that models trained under zero-shot conditions significantly underperform compared to fully supervised models [9].

D. Class Imbalance and Metric Interpretation

Class imbalance is an issue that is very prevalent among NIDS datasets and can render the accuracy metric quite misleading [18]. More relevant indicators of the performance are the recall, precision, F1-score, and the false-positive rate. For zero-day detection, in particular, high recall for unseen classes is a primary goal, even at the cost of some false positives. Studies should report metrics for both seen and unseen categories separately to provide a clearer picture of generalization performance.

E. Explainability and Real-World Deployment

The complex AI model used for suspicious traffic detection might not give any clear explanations for its conclusions. Explainable AI can help to make intrusion detection results more usable by security analysts [8]. Other factors that can impact reliability in the context of practical application include the changing nature of traffic, concept drift, delayed labeling, class imbalance, and computational constraints. In summary, zero-day NIDS evaluation requires an understanding of all these aspects together, rather than an independent consideration of the model performance.

VII. RESEARCH GAPS AND FUTURE DIRECTIONS

The reviewed literature highlights several areas that require further attention for reliable zero-day intrusion detection.

A. Standardized Zero-Day Evaluation

There is no unified evaluation protocol that can be used to characterize the zero-day attack. Future research needs to clarify the set of information that will be used at the training and testing stages when addressing zero-day situations. Standardized benchmarks that combine attack-family holdout, temporal splits, and cross-dataset evaluation would facilitate fairer comparison across studies. A community-agreed-upon evaluation framework analogous to those in vision or NLP benchmarks could significantly advance the field.

B. Cross-Dataset and Temporal Generalization

Most of the existing studies use individual benchmark dataset, which makes it impossible to draw any conclusions regarding the performance of the system in different networking environment. Cross-dataset experiments and temporally structured evaluations that incorporate concept drift would be essential next steps. Developing datasets with richer temporal metadata and more diverse network environments would facilitate such studies.

C. Adaptive and Few-Shot Learning

The pattern of attacks can vary depending on the time period, making it hard to update the model. Adaptive, continual and few-shot learning approaches can serve as potential directions for model updates with small amounts of new data. Meta-learning frameworks that can quickly adapt to novel attack patterns with minimal labeled examples show particular promise for practical zero-day detection deployments.

D. Explainability and Deployment-Aware Evaluation

In the future, a zero-day attack detection system must be able to present interpretable evidence of the threat so that security analysts could check detected threats. The evaluation should also take into account such practical considerations as false positives, computational complexity, changes in traffic, and delayed labeling of the attacks.

E. Improved Dataset Practices

Future research will be facilitated by improved documentation of datasets and the clear description of how attacks were generated, traffic behavior, pre-processing techniques, and duplicate/very similar record handling. Diverse, realistic, and temporally extended datasets that cover emerging attack types and network environments are needed to support robust zero-day NIDS evaluation.

Overall, future research on zero-day NIDS should evolve away from simply achieving high accuracy to evaluation that is standardized and generalizable.

VIII. CONCLUSION

In this review, AI methods for detecting zero-day attacks in network environments were analyzed with special focus on detection methods, datasets, evaluation criteria, and research problems. From the analysis of 20 recent papers, it is found that the research landscape is dominated by supervised models applied to standard benchmark datasets with conventional train-test splits. This configuration, while producing high accuracy scores, frequently overestimates the ability of a model to detect genuinely unseen attacks.

For example, the use of standard train-test split does not allow demonstrating the ability of models to generalize over zero-days. Evaluation can be done using attack-family hold-out, temporal, and cross-dataset approaches, which are more robust but less frequently applied. Dataset-specific findings, class imbalance, and lack of explainability add additional complications to real-world deployment.

Therefore, future research should focus on developing standard methods of evaluation of zero-day detection, cross-dataset and temporal verification of algorithms, adaptive learning techniques, explainable AI techniques, and the improvement of dataset quality. Moving toward evaluation-centric research will help build NIDS that are not only effective in controlled benchmarks but also reliable against the unpredictable nature of real-world zero-day threats.

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